Joel Oskarsson

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Research Interests

My research is focused on probabilistic machine learning methods for modeling data with spatial and temporal dependencies. I am motivated by applications of machine learning to Earth system modeling, such as weather forecasting and climate modeling. Part of my research is in core machine learning method development, where unique challenges from Earth science act as an underlying motivation. These challenges include: irregular observations, high-dimensional data, complex temporal dynamics and accurate uncertainty estimatation. I also enjoy developing methods specifically for Earth science applications, tailoring the machine learning methods using domain knowledge to solve relevant problems in new ways.

Employment and Education

2025 – **Postdoctoral Fellow, ETH Zurich AI Center**
Zurich, Switzerland

- Researching machine learning for earth system modeling.
- Awarded 2-year fellowship in highly competitive call with a 4% acceptance rate.
- Mentored by Professors Torsten Hoefler, Siddhartha Mishra, and Benedikt Soja.
- Fellowship supported by the Swiss AI Initiative, part of effort to develop AI foundation models.

2020–2025 **PhD in Computer Science**
Linköping University, 240 ECTS
Linköping, Sweden

- PhD thesis: *Modeling Spatio-Temporal Systems with Graph-based Machine Learning*.
- Employed during full period, with teaching duties making up 20% of work activities.
- At the Division of Statistics and Machine Learning (STIMA), Department of Computer and Information Science. Supervised by Fredrik Lindsten (main supervisor), Per Sidén and Tomas Landelius.
- Affiliated PhD Student in the Wallenberg AI, Autonomous Systems and Software Program (WASP).

2015–2020 **Master's program in Computer Science and Engineering**
Linköping University, 300 ECTS
Linköping, Sweden

- Master's thesis: *Probabilistic Regression using Conditional Generative Adversarial Networks*.
- Exchange studies at ETH Zurich, Switzerland during first year of master's, 2018–2019.

Additional Experience

2016 – 2019 **Teaching Assistant, Linköping University**
Linköping, Sweden

2018, Summer **Summer Intern, Ericsson Research**
Linköping, Sweden

Publications

- 6 conference papers, 2 journal papers, 7 workshop papers, + 3 works under review.
- Main publication venues include top machine learning conferences, as well as subject-specific journals.
- 403 total citations, h-index 9 (from Google Scholar, October 2026).

Journal Papers

[1] Vignesh Gopakumar, Ander Gray, **Joel Oskarsson**, Lorenzo Zanisi, Daniel Giles, Matt J Kusner, Stanislas Pamela, and Marc Peter Deisenroth. "Uncertainty quantification of surrogate models using conformal prediction". In: *Machine Learning: Science and Technology* (2026).

[2] Theodor Westny, **Joel Oskarsson**, Björn Olofsson, and Erik Frisk. "MTP-GO: Graph-Based Probabilistic Multi-Agent Trajectory Prediction with Neural ODEs". In: *IEEE Transactions on Intelligent Vehicles* (2023).

Peer-Reviewed Conference Papers

[3] Daniel Holmberg, **Joel Oskarsson**, Erik Wikingsson, Fredrik Lindsten, and Teemu Roos. "Njord: A Probabilistic Graph Neural Network for Ensemble Ocean Forecasting". In: *Advances in Neural Information Processing Systems*. 2026.

[4] Martin Andrae, Tomas Landelius, **Joel Oskarsson**, and Fredrik Lindsten. "Continuous Ensemble Weather Forecasting with Diffusion models". In: *International Conference on Learning Representations*. 2025.

[5] **Joel Oskarsson**, Tomas Landelius, Marc Peter Deisenroth, and Fredrik Lindsten. "Probabilistic Weather Forecasting with Hierarchical Graph Neural Networks". In: *Advances in Neural Information Processing Systems*. Vol. 37. [**Spotlight**]. 2024.

[6] **Joel Oskarsson**, Per Sidén, and Fredrik Lindsten. "Temporal Graph Neural Networks for Irregular Data". In: *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*. 2023.

[7] Theodor Westny, **Joel Oskarsson**, Björn Olofsson, and Erik Frisk. "Evaluation of Differentially Constrained Motion Models for Graph-Based Trajectory Prediction". In: *2023 IEEE Intelligent Vehicles Symposium (IV)*. 2023.

[8] **Joel Oskarsson**, Per Sidén, and Fredrik Lindsten. "Scalable Deep Gaussian Markov Random Fields for General Graphs". In: *Proceedings of the 39th International Conference on Machine Learning*. PMLR, 2022.

Peer-Reviewed Workshop Papers

[9] Younes Elberkennou, Dmitri Demler, Thierry Meier, Luca Rispoli, Fanny Lehmann, and **Joel Oskarsson**. "Every Fixed Metric Has a Blind Spot: A Learned Atmospheric Critic for Scoring Forecast Realism". In: *NeurIPS 2026 Workshop on Advances in Representation Learning for Earth Observation*. 2026.

[10] Alessandro Lucatelli, Ferdinand Unterhuber, Katharina Krenzer, Menna El-Assady, Benedikt Soja, Christina Humer, and **Joel Oskarsson**. "GNN-VIEW: Visual Inspection and Explanation of Graph-based Earth-System Models". In: *NeurIPS 2026 Workshop on Advances in Representation Learning for Earth Observation*. 2026.

[11] Erik Larsson, **Joel Oskarsson**, Tomas Landelius, and Fredrik Lindsten. "CRPS-LAM: Regional ensemble weather forecasting from matching marginals". In: *EurIPS 2025 Workshop on AI for Climate and Conservation*. 2025.

[12] Erik Larsson, **Joel Oskarsson**, Tomas Landelius, and Fredrik Lindsten. "Diffusion-LAM: Probabilistic Limited Area Weather Forecasting with Diffusion". In: *ICLR 2025 Workshop on Tackling Climate Change with Machine Learning*. 2025.

[13] Vignesh Gopakumar, **Joel Oskarsson**, Ander Gray, Lorenzo Zanisi, Stanislas Pamela, Daniel Giles, Matt Kusner, and Marc Deisenroth. "Valid Error Bars for Neural Weather Models using Conformal Prediction". In: *ICML Workshop on Machine Learning for Earth System Modeling*. 2024.

[14] **Joel Oskarsson**, Tomas Landelius, and Fredrik Lindsten. "Graph-based Neural Weather Prediction for Limited Area Modeling". In: *NeurIPS 2023 Workshop on Tackling Climate Change with Machine Learning*. 2023.

[15] **Joel Oskarsson**, Per Sidén, and Fredrik Lindsten. "Temporal Graph Neural Networks with Time-Continuous Latent States". In: *ICML Workshop on Continuous Time Methods for Machine Learning*. 2022.

Under Review

[16] Simon Adamov*, **Joel Oskarsson***, Leif Denby, Tomas Landelius, Kasper Hintz, Simon Christiansen, Irene Schicker, Carlos Osuna, Fredrik Lindsten, Oliver Fuhrer, and Sebastian Schemm. "Building Machine Learning Limited Area Models: Kilometer-Scale Weather Forecasting in Realistic Settings". In: *preprint* (2025).

*Equal contribution.

[17] Erik Wikingsson, **Joel Oskarsson**, Tomas Landelius, and Fredrik Lindsten. "CRPS-LAM: Probabilistic Regional Weather Forecasting with Continuous Ranked Probability Score". In: *preprint* (2026).

[18] Jiakai Chen, **Joel Oskarsson**, Simon Driscoll, and Sebastian Schemm. "Butterfly Effect and the Kinetic Energy Cascade in Probabilistic Machine Learning Weather Prediction Models". In: *preprint* (2026).

Awarded Grants

Co-Applicant

2026 ICAIN-IGAIP project (320,000 CHF ≈ 345,000 EUR, 4/40 applications funded)
Project *AI-Powered Early Warning and Profiling for Displaced Populations*, headed by Christina Humer (PI, ETHZ) and in collaboration with the International Organization for Migration.

Compute Grants

2026 Co-applicant for Swiss-AI Large compute grant (754,000 total GPUh)
Head of work package *Efficient and Distributionally Accurate Generative Ensemble Models* (60,000 GPUh)

Career Development Grants

2023 Research Visit Scholarship (60,000 SEK ≈ 5,200 EUR)
Wallenberg AI, Autonomous Systems and Software Program (WASP)

Invited Talks

2026-09-24 *GNNs for Spatio-Temporal and Earth System Modeling*
KCDS Summer School on Graph Neural Networks, Karlsruhe Institute of Technology (KIT). Karlsruhe, Germany

2026-02-16 *Probabilistic Deep Learning for Earth System Modeling*
WeatherGenerator Dissemination Workshop. Oslo, Norway

2025-05-02 *Probabilistic Weather Forecasting with Hierarchical Graph Neural Networks*
Google DeepMind. Online

2024-11-07 *Frontiers in Machine Learning for Weather Forecasting*
RISE Research Institutes of Sweden, Learning Machines Seminars. Online

2024-08-22 *Graph-based Machine Learning for Weather*
European Centre for Medium-Range Weather Forecasts (ECMWF). Bonn, Germany

2024-05-02 *Graph-based Neural Weather Prediction for Limited Area Modeling*
MeteoSwiss. Zurich, Switzerland

2024-04-30 *Graph-based Neural Weather Prediction for Limited Area Modeling*
ETH Zurich. Zurich, Switzerland

2024-04-17 *Graph-based Machine Learning for Spatio-Temporal Data, with Application to Traffic and Weather Forecasting*
University of New South Wales (UNSW). Sydney, Australia (Online)

2024-03-05 *Graph-based Neural Weather Prediction for Limited Area Modeling*
Uppsala University. Uppsala, Sweden

2024-01-11 *Neural Weather Prediction for Limited Area Modeling*
IEA Wind Task 51 Webinar: Forecasting for the Weather Driven Energy System. Online

2023-10-10 *Graph-based Neural Weather Prediction for Limited Area Modeling*
Danish Meteorological Institute. Copenhagen, Denmark

Research Visits

Longer Visits

2023 University College London (UCL)
London, U.K.
Visited the Sustainability and Machine Learning Group at UCL for 5 weeks. Hosted by Marc Deisenroth.

Shorter Visits ($\leq$ 1 Week)

2026 Norwegian Meteorological Institute
Oslo, Norway. Hosted by Thomas Nipen

2024 European Centre for Medium-Range Weather Forecasts (ECMWF)
Bonn, Germany. Hosted by Mariana Clare

2024 ETH Zurich + MeteoSwiss
Zurich, Switzerland. Hosted by Simon Adamov

2022 Finnish Center for AI + Aalto University
Helsinki, Finland. WASP International Study Visit

Academic Service and Community Activities

2023– Creator and maintainer of the *Neural-LAM* open source python library for machine learning weather forecasting models.

- Over 300 GitHub forks and stars.
- Used by multiple international research groups as basis for peer-reviewed publications.
- Mentor for related Google Summer of Code project 2026.

2026 Co-organizer of the *AI + Environment Summit 2026*, a one-day workshop on AI in environmental sciences in Zurich, Switzerland.

- Responsible for session on AI for weather and climate, and poster session.

2023–2025 Co-organizer of monthly online meetups for researchers working on machine learning limited area weather forecasting models.

2022–2025 Organizer of the *STIMA Machine Learning Reading Group* at Linköping University.

Reviewing and Editorial Service

Area Chair

2027 International Conference on Learning Representations (ICLR)

Reviewer

2024,2026* Conference on Neural Information Processing Systems (NeurIPS)

2025,2026 International Conference on Learning Representations (ICLR)

2023,2024,2026 International Conference on Artificial Intelligence and Statistics (AISTATS)

2026 Geoscientific Model Development (GMD)

2026 Quarterly Journal of the Royal Meteorological Society (QJRMS)

2026 Climate Informatics conference

2026 Geomatics, Natural Hazards and Risk

2025 Geophysical Research Letters (GRL)

2024,2025 Tackling Climate Change with Machine Learning Workshop

2025 Scientific Online Letters on the Atmosphere (SOLA)

Supervision and Teaching

PhD Students (Co-supervisor)

- Faro Schaefer (ETHZ, starting December 2026).

Thesis Supervision

- 10 master's theses (8 LiU and 2 ETHZ, 1 ongoing).
- 1 bachelor's thesis (ETHZ, ongoing).
- 2 semester theses (ETHZ).

*Recognized as a "Top Reviewer".

Teaching

- Guest lecturer for courses *AI Center Projects in Machine Learning Research* (ETHZ) and *Time Series and Sequence Learning* (LiU).
- Co-developer of the online course *Foundations of Machine Learning* (LiU, 2021–2024).
- Pedagogical training: *Becoming a Teacher in Higher Education*, 6 ECTS (LiU, 2021).
- Teaching assistant for courses in machine learning (introductory and advanced), programming and mathematics (LiU, 2016–2025).

Scientific Outreach and Popular Science

2026 *Towards Regional High-Resolution Weather Forecasting with ML*
Contributed to panel discussion as part of the Destination Earth course *Machine Learning for Earth Systems Modelling*.

2025 *AI för prognosmeteorologer*
Lecture for professional forecast meteorologists in Sweden, about general AI technology and AI models for weather forecasting.

2025 *Forskningsframsteg inom Maskininlärning för Väderprognoser*
Popular science article about machine learning weather forecasting published in Polarfront, the journal of the Swedish meteorological society.

2025 *Doing a PhD in Machine Learning and AI within WASP*
Talk for undergraduate students about machine learning research and doing a PhD within the AI area.

Media Coverage of Research

2024 *De vill göra väderprognoser mer pålitliga – med hjälp av AI*
News coverage, SVT (Swedish Public Service Television)

2023 *Förbättrade väderprognoser med AI*
News posting, Linköping University

Participation in Scientific Events

2026 *Climate Informatics*
Lausanne, Switzerland
Oral presentation

2025 *Royal Swedish Academy Of Sciences Symposium on AI for Science*
Stockholm, Sweden
Oral presentation

2025 *Workshop on Machine Learning for Earth System Modelling*
Bonn, Germany
Oral presentation

2025 *Learning on Graphs and Geometry Meetup*
Uppsala, Sweden
Oral presentation

2024 *Conference on Neural Information Processing Systems (NeurIPS)*
Vancouver, Canada
Poster presentation

2024 *ELLIIT Focus Period Symposium: Machine Learning for Climate Science*
Linköping, Sweden
Poster presentation

2024 *Large-Scale Deep Learning for the Earth System Workshop*
Bonn, Germany
Oral presentation

2024 *International Conference on Machine Learning (ICML)*
Machine Learning for Earth System Modeling Workshop at ICML 2024
Vienna, Austria
Workshop poster presentation

2024 *ESA-ECMWF workshop on Machine Learning for Earth System Observation and Prediction*
Frascati, Italy
Oral presentation + poster presentation + panel discussion

2024 *ELLIIT Annual Workshop*
Lund, Sweden
Poster presentation

2024 *WASP Winter Conference*
Norrköping, Sweden
Poster presentation

2023 *Conference on Neural Information Processing Systems (NeurIPS)*
Tackling Climate Change with Machine Learning Workshop at NeurIPS 2023
Online attendance
Workshop poster presentation

2023 *Large-Scale Deep Learning for the Earth System Workshop*
Bonn, Germany
Oral presentation

2023 *Nordic Conference in Mathematical Statistics (NORDSTAT)*
Gothenburg, Sweden
Oral presentation

2023 *International Conference on Artificial Intelligence and Statistics (AISTATS)*
Valencia, Spain
Poster presentation

2023 *Swedish Symposium on Image Analysis and Deep Learning*
Norrköping, Sweden
Poster presentation

2023 *WASP Winter Conference*
Norrköping, Sweden
Poster presentation

2022 *AI Day Finland*
Helsinki, Finland

2022 *Learning on Graphs Conference (LoG)*
Online attendance

2022 *International Conference on Machine Learning (ICML)*
Continuous Time Methods for Machine Learning Workshop at ICML 2022
Baltimore, U.S.
Poster presentation + Workshop poster presentation (*Cancelled due to illness*)

2022 *Nordic Probabilistic AI Summer School*
Helsinki, Finland
Poster presentation

Specific Skills and Knowledge

- Extensive knowledge of modern **machine learning** methods. Specific expertise in:
  - Probabilistic deep learning
  - Spatio-temporal models of physical systems
  - Bayesian models and inference methods
  - Graph neural networks
- Experience with high-performance computing: Planning, applying for compute grants, and making efficient use of allocated resources.
- Languages:
  - Swedish: Native
  - English: Fluent
  - German: Basic skills (A2)

References

- *Prof. Siddhartha Mishra*

  Position: Professor of Applied Mathematics at ETH Zurich, Switzerland

  Relation: Postdoc advisor

  E-mail: `siddhartha.mishra@sam.math.ethz.ch`

- *Prof. Torsten Hoefler*

  Position: Professor of Computer Science at ETH Zurich, Switzerland

  Relation: Postdoc advisor

  E-mail: htor@inf.ethz.ch

- *Prof. Fredrik Lindsten*

  Position: Professor of Machine Learning at Linköping University, Sweden

  Relation: PhD advisor

  E-mail: fredrik.lindsten@liu.se

- *Dr. Leif Denby*

  Position: Senior Scientist at the Danish Meteorological Institute, Denmark

  Relation: Research collaborator

  E-mail: lcd@dmi.dk